Clearboy hammer bubbler

Hand-blown boro 3.3 · 140 mm · one-piece head, stem and foot

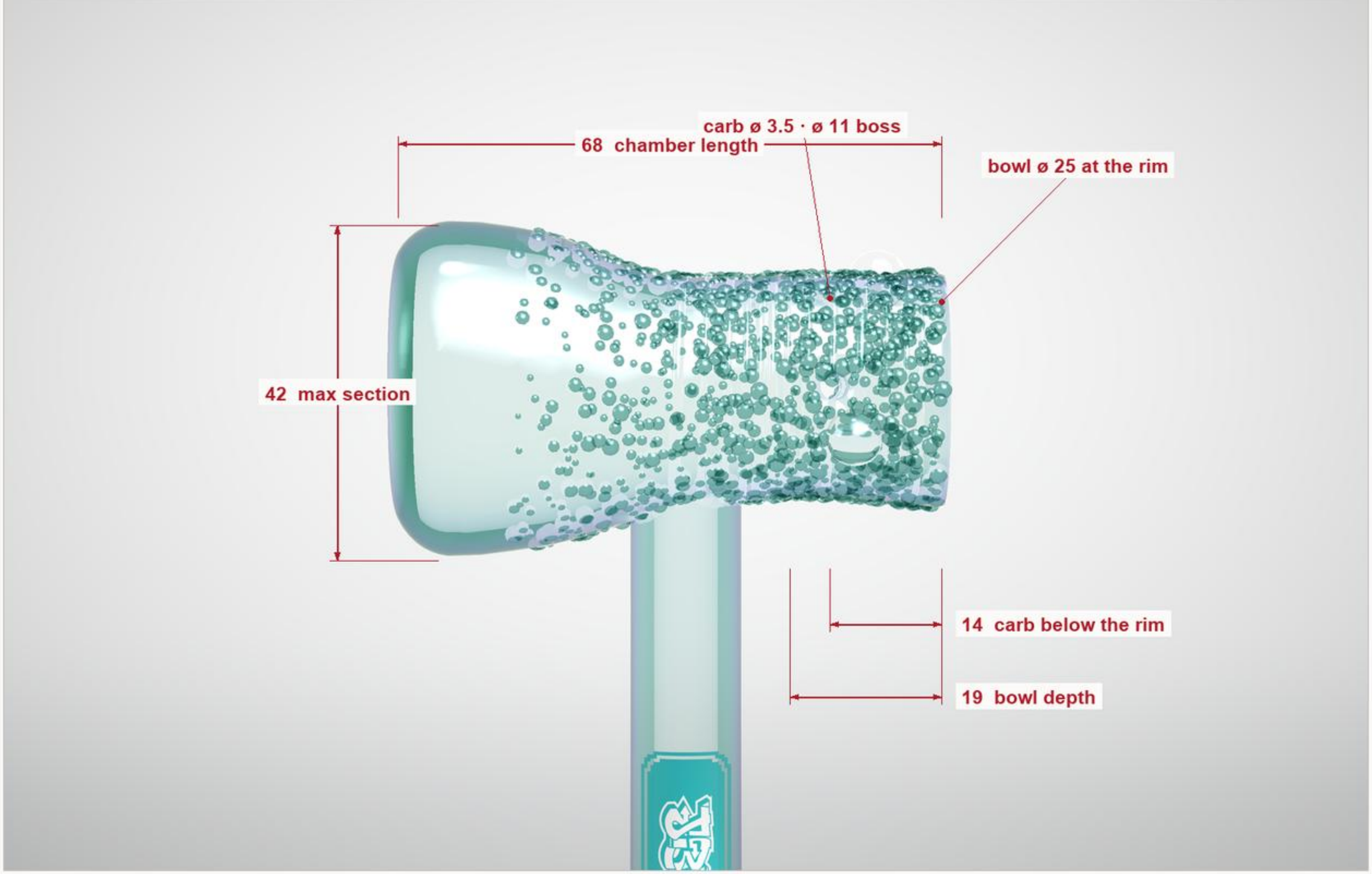
OVERALL		
Height, standing on the foot	**140 mm**	5.51 in
Length, laid down	**140 mm**	
Glass	**Borosilicate 3.3**	clear stock, fumed
Finished mass	**≈ 81 g**	from the solid model
HEAD		
Chamber length	**68 mm**	62–71 across views — hand-shaped
Max section	**42 × 37 mm**	oval, not a cylinder
Wall	**≈ 3 mm**	**INFERRED — confirm before tooling**
STEM		
Tube OD	**ø 14 mm**	**original measured ø 11 — thickened for the label**
Bore ID	**ø 8 mm**	wall ≈ 3 mm
Exposed length	**88 mm**	foot face to the head
Foot / mouthpiece disc	**ø 24.5 × 7 mm**	edges broken 1.6 mm
BOWL		
Opening ID at the rim	**ø 25 mm**	
Depth, rim to the hole	**19 mm**	
Hole in the bottom of the bowl	**ø 3 mm**	**original measured ø 5**
Carb hole	**ø 3.5 mm**	on a ø 11 boss, 14 mm below the rim

Head and bowl

Chamber length, section, bowl opening and the carb, taken off the solid model.

Head and bowl

Broadside. The chamber is a shaped solid, not a cylinder — the section is oval and varies along the axis.

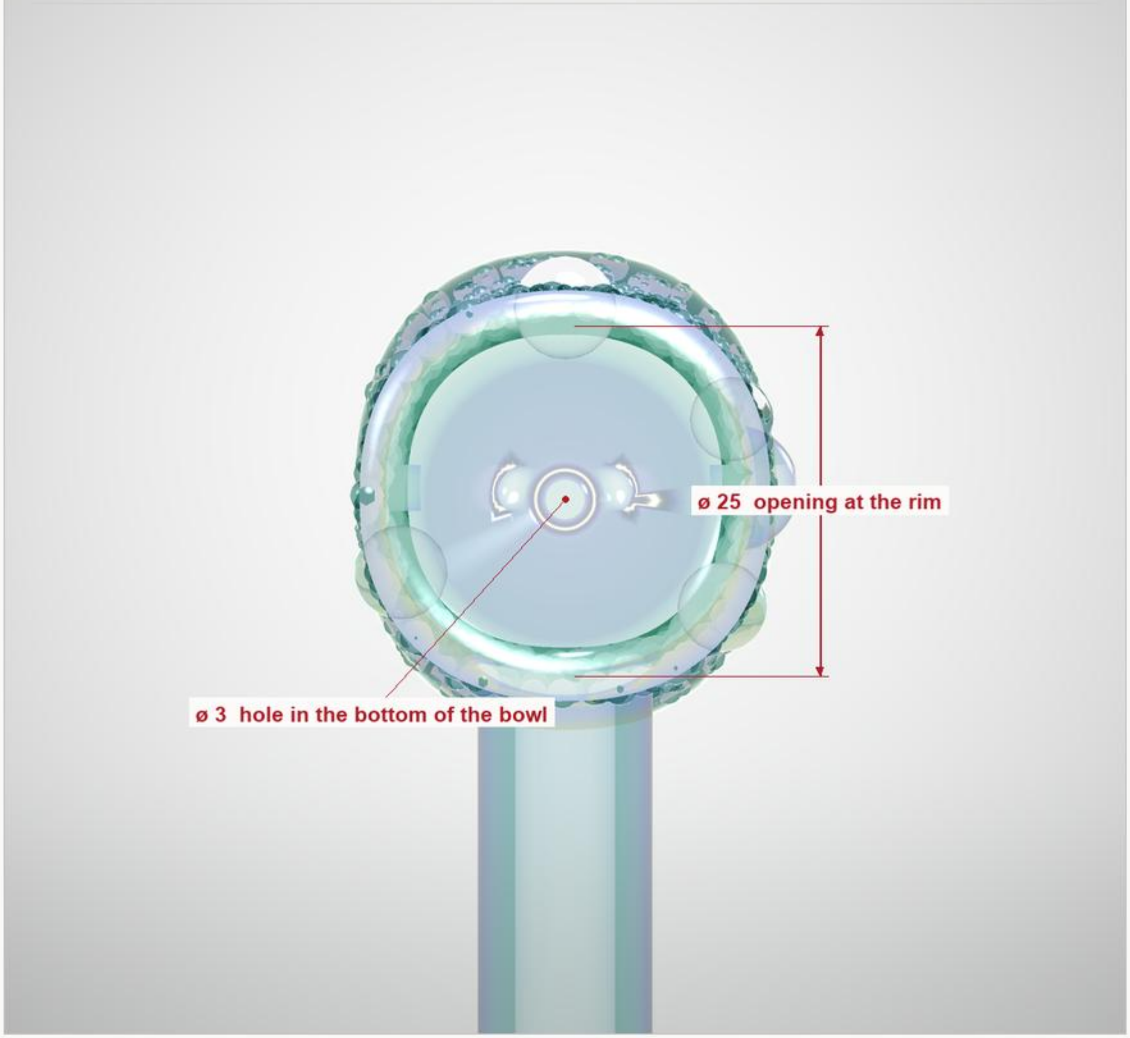


Bowl end

Looking straight down the head axis at the opening and the hole beneath it.

Bowl end, on the head axis

The hole in the bottom of the bowl is called down to ø 3. The original measured ø 5 — do not correct it back.

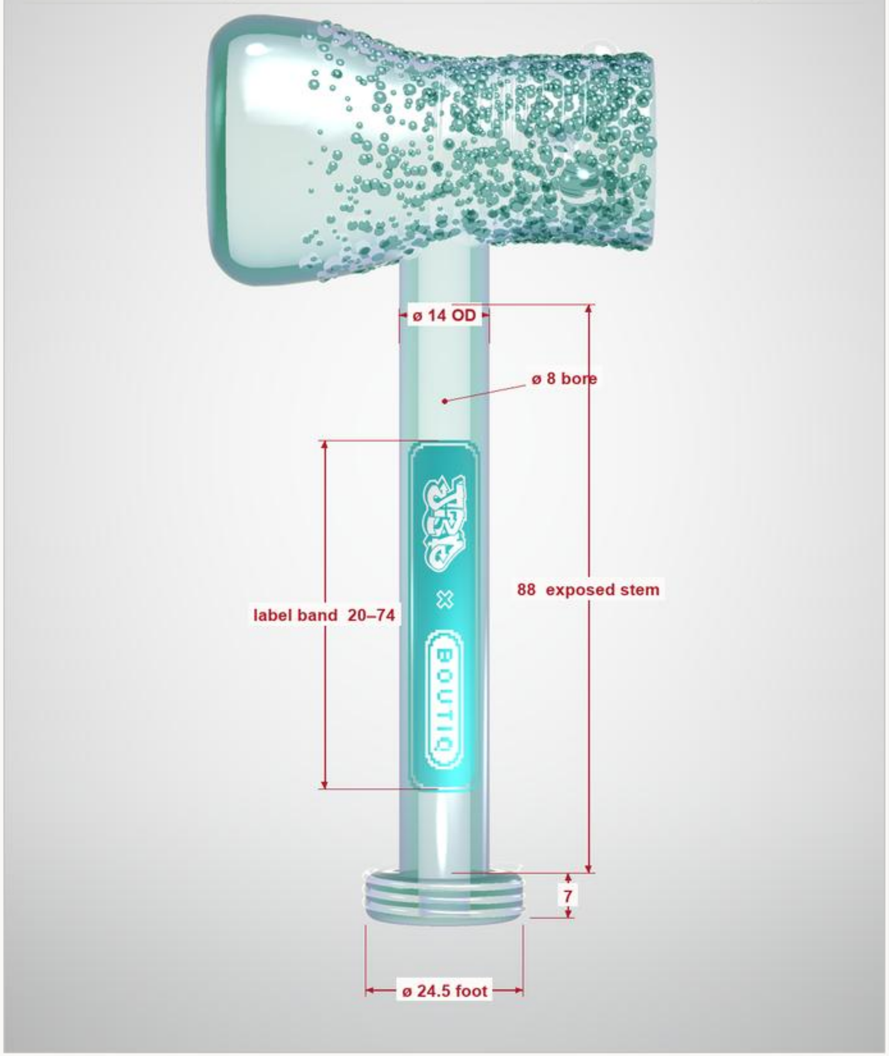


Stem, foot and label band

Where the enamel band sits, and the disc the piece stands on.

Stem, foot and label band

Stem thickened from the original ø 11 so the enamel band has a flat to sit on. The bore runs the full length.

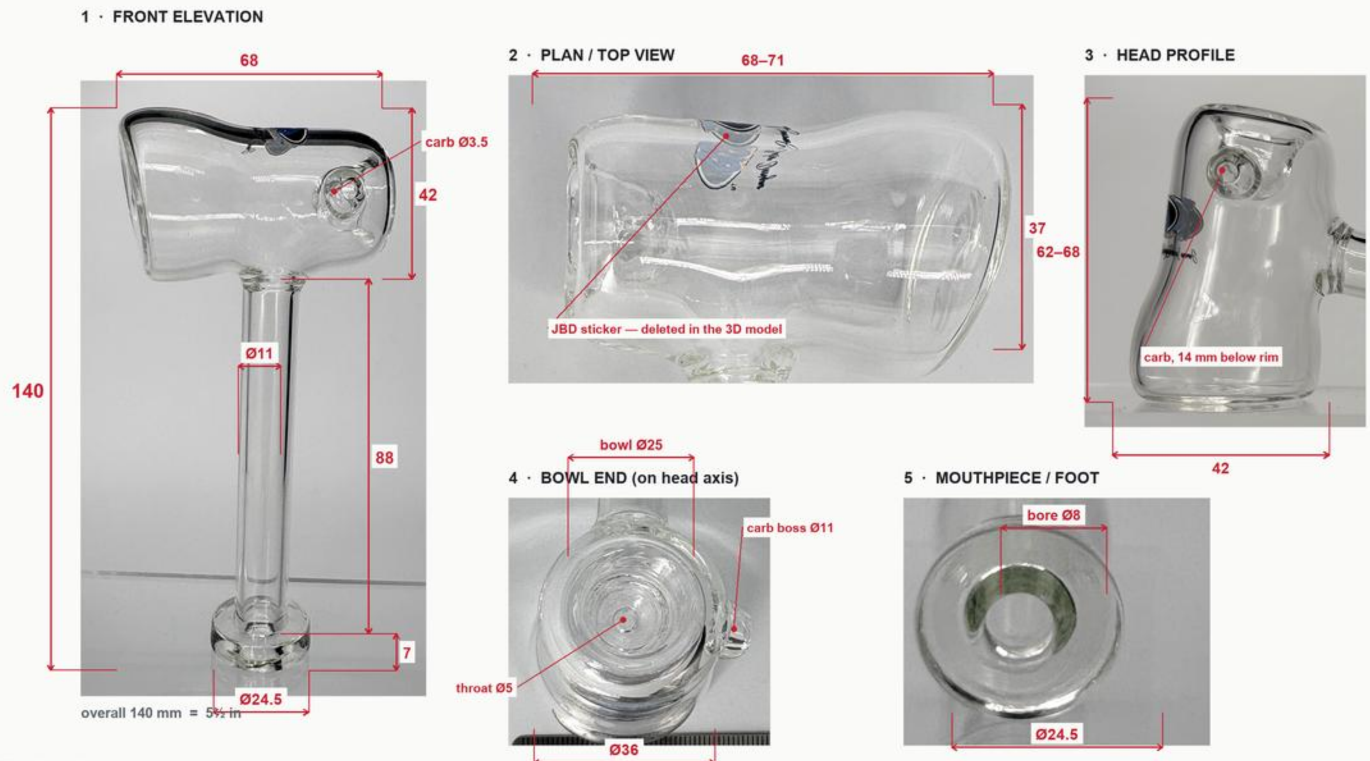


Dimensional survey

The original piece, measured photogrammetrically against a stainless rule — the reference the build was drawn from

JEROME BAKER DESIGNS · “CLEARBOY” HAMMER — DIMENSIONAL SURVEY

Measured photogrammetrically from the ruler-referenced photos (IMG_5850–5859). All figures in millimetres, ±2–3 mm. Hand-blown — the head is not axisymmetric.



MEASURED SCHEDULE

OVERALL		STEM & FOOT		BOWL & CARB	
Overall height, standing on foot	140 mm (5.51 in)	Stem tube OD	11 mm	Bowl opening ID at rim	25 mm
Overall length, laid down	140 mm	Stem bore ID	8 mm (wall ≈ 1.6 mm)	Bowl throat / drop hole	5 mm
Head (chamber) length	68 mm (62–71 range across views)	Exposed stem length	88 mm	Bowl depth to throat	18–20 mm
Head max section	42 × 37 mm (oval, hand-shaped)	Foot / mouthpiece disc	Ø24.5 × 7 mm thick	Carb hole · boss · position	Ø3.5 · Ø11 · 14 mm below rim

Method: scale set from the stainless rule in IMG_5855–5859 (mm graduations resolved to ±0.5 px/mm), cross-checked across four independent set-ups and corrected for stand-off parallax.
Head length and section vary 5–10% between views because the piece is hand-blown and slightly oval — treat the head as a shaped solid, not a cylinder. Confirm with calipers before any tooling.

Nug jar and cork

Straight cylinder, flat closed bottom, tapered natural cork · 92 mm glass

BODY

Glass height, rim to bench	**92 mm**	
Outside diameter	**ø 44 mm**	straight cylinder, no shoulder
Wall	**3 mm**	
Floor	**3 mm**	flat, closed
Glass	**Borosilicate 3.3**	clear stock, fumed
Finished mass	**≈ 90 g**	glass only

MOUTH

Opening ID	**ø 38 mm**	the spec the cork is cut to

CORK

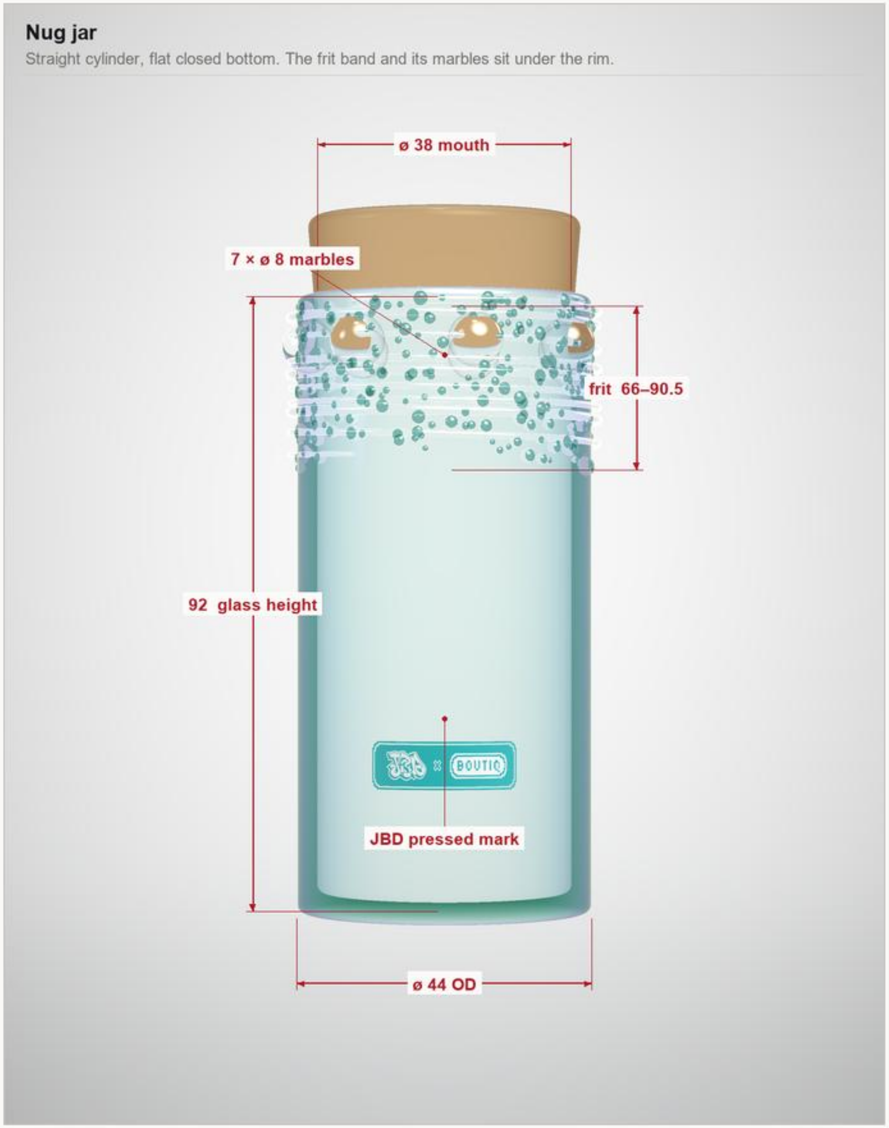
Overall length	**27 mm**	natural cork, no mushroom cap
Diameter, bottom / top	**ø 36.6 / ø 41 mm**	seats on the taper
Seat depth in the mouth	**15 mm**	≈ 12 mm stands proud
Mass	**≈ 10 g**	

NOTES

1. Hand-blown. The figures above are nominal targets, not machined tolerances — the head is deliberately oval and varies 5–10% between views on the original.
2. Hold these four: bowl opening ø 25, hole in the bottom of the bowl ø 3, stem bore ø 8, jar mouth ø 38. Everything else can move with the glass.
3. Chamber wall is inferred, not measured. Two caliper readings on the original — rim and stem OD — lock the whole model; mass, volume and glass cost all move if they come back different.
4. Supplied with this sheet: clearboy_hammer.step / .stl and jar.step / .stl, plus the frit, marble and cork meshes. STEP is the reference; the meshes are for print and mould work only.

Nug jar elevation

Body, mouth, frit band and the marbles set around the opening.

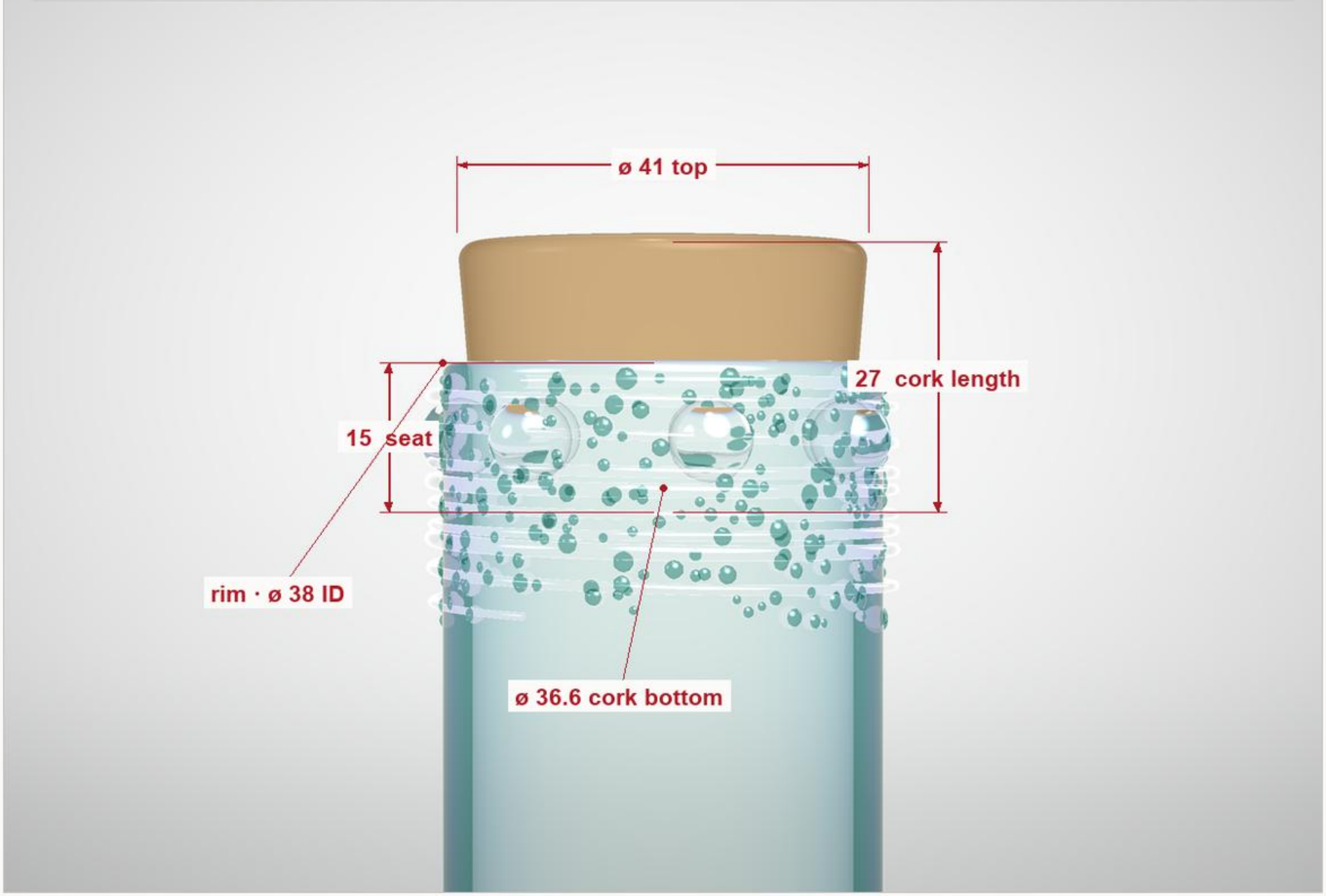


Mouth and cork

The taper the cork seats on, and how far it stands proud.

Mouth and cork

One gently tapered plug, no mushroom cap. It seats on the taper, about 12 mm standing proud.



Decoration and colourways

Frit, marbles, the enamel label and the pressed mark — both colourways carry the same work

HAMMER

Frit	Rolled over the bowl end of the head — the outer 54 mm of the 68 mm chamber, densest at the rim. Worked in, not a coating.
Linework	About 13 turns at 2.4 pitch over the head, plus rings at the foot. Clear over the fritted body. Laid down before the marbles, so it runs behind them.
Marbles	4 clear marbles, ø 6.5–7.5, set into the fritted band over the linework. Hand-placed, not evenly spaced.
Label	Enamel, on the stem, 20–74 mm up from the foot face. JBD × Boutiq dropped out in white. The print reads one way along the stem — strike it to read with the head to the left.

JAR

Frit	Band around the opening, 66–90.5 mm up from the base.
Linework	9 turns at 2.4 pitch over the frit band, laid down before the marbles.
Marbles	7 clear marbles, ø 8, evenly spaced, centres 84.5 mm up, set over the linework.
Mark	JBD pressed into a molten stamp pad, lower middle, centre 28 mm up. Die face ≈ 28 × 11 mm; the mark itself runs ≈ 25 mm wide.

COLOURWAYS

Bluish teal · silver fume	Bluish teal body, silver fume. Teal frit, clear marbles.
Magenta · gold fume	Magenta body, gold fume. Magenta frit, clear marbles.

Process — hammer bubbler

Bench sequence. The shop's working order governs where it differs — the numbered checks do not move.

STOCK

1 Borosilicate 3.3 throughout. Chamber off tube nominally ø 38-42 with a 3.5-4 wall; stem off ø 14 OD / ø 8 ID tube; foot off the same stem stock.

2 Colour: transparent teal or magenta rod for the wash, matched frit, clear marble stock, and silver or gold for the fume per colourway.

CHAMBER

3 Close and shape one end into the rounded lobe. Blow the chamber out to 68 long, working the section oval to 42 × 37 - paddle and marver it, do not blow it round in a mould.

4 Fume the inside before any colour goes on. The fume is what shifts in use; colour laid under it kills that.

5 Lay the transparent wash over the fume, even from lobe to rim.

FRIT AND LINEWORK

6 Roll the bowl end in frit over the outer 54 mm of the chamber, densest at the rim, thinning out toward the lobe. Melt it in flush - it is worked into the wall, not a coating sitting on it.

7 Spin the linework down next, about 13 turns at 2.4 pitch over the head, plus a few rings round the foot. Clear over the fritted body.

MARBLES

8 Set 4 clear marbles ø 6.5-7.5 into the fritted band by hand, on top of the linework - the lines run behind them, not across them.

9 They are not evenly spaced and should not look it. Encase them fully; no open seam.

BOWL AND CARB

10 Open the bowl at the rim end to ø 25, 19 deep, funnelled down to the ø 3 hole in the bottom of the bowl. Hold the ø 3 - the original measured ø 5 and it was called down on purpose.

11 Raise the ø 11 carb boss on the side wall 14 mm below the rim, then open the ø 3.5 carb through it.

STEM AND FOOT

12 Cut the stem and fuse it into the underside of the chamber at mid-length. The ø 8 bore must break through into the chamber clean - no web, no pinch.

13 Flare the foot to ø 24.5 × 7 thick and break the edges about 1.6. It has to stand square: 140 overall, no rock.

Process — nug jar and cork

Same order of work: shape, fume, colour, frit, marbles, mark.

BODY

1 Cut ø 44 × 3 wall tube to 92 plus working allowance. Close the bottom flat with a 3 floor - flat enough to stand without rocking.

2 Fume the inside, then lay the wash over it, same order as the hammer.

FRIT AND LINEWORK

3 Frit band 66-90.5 up from the base, melted in flush.

4 Spin 9 turns at 2.4 pitch over the frit band. Lines go down before any marble is set.

MARBLES AND MARK

5 Set 7 clear marbles ø 8 evenly round the opening, centres 84.5 up, over the linework. Evenly spaced here - the jar reads as a ring, the hammer does not.

6 Press the JBD mark into a molten stamp pad on the wall, lower middle, centre 28 up. Die face about 28 × 11.

MOUTH

7 Open and true the mouth to ø 38 ID. Hold the OD at 44 right up to the rim - straight cylinder, no shoulder, no flare.

CORK

8 Natural cork, 27 long, ø 36.6 at the bottom and ø 41 at the top. One gentle taper, no mushroom cap.

9 Fit to the piece it ships with: 15 into the mouth, about 12 standing proud, seating on the taper rather than bottoming out.

Anneal, decoration and QC

What happens after the torch, and what gets measured before a piece is passed.

ANNEAL

1 Anneal every piece. Typical for boro 3.3 is a soak around 565 °C / 1050 °F held 30-60 minutes by mass, then down to about 480 °C at 50 °C/hour before free cooling - use the shop's own kiln schedule where it differs.

2 Check for strain under a polariscope after cooling. The stem-to-chamber joint and the marble seats are where it shows first.

DECORATION AFTER ANNEAL

3 The enamel label goes on the stem after annealing: 20-74 mm up from the foot face, JBD × Boutiq dropped out in white, kiln-cured to the glass.

4 Strike the print to read with the head to the left. The box lays the piece that way; struck the other way round it reads upside down in the tray.

HOLD POINTS

5 H1 chamber - 68 long, 42 × 37 section, before the bowl is opened.

6 H2 bowl - ø 25 at the rim, 19 deep, ø 3 hole on a go / no-go pin.

7 H3 stem - blow through end to end; the ø 8 bore must be clear.

8 H4 after anneal - 140 overall, ø 14 stem, ø 24.5 × 7 foot, stands without rocking, no strain.

9 H5 jar - ø 38 mouth on a go / no-go, cork seats 15 with about 12 proud, stands flat.

10 H6 finish - no chips at the rim or the carb, marbles fully encased, label square to the stem and centred on the tube.

PACKING

11 Into the Boutiq tray: hammer head to the left in the pipe recess so the label reads left to right, jar with the cork end to the back-left of its trough.

12 Reject rather than pass anything failing H2, H3 or H4. The four dimensions that must hold are bowl ø 25, bowl hole ø 3, stem bore ø 8, jar mouth ø 38.